

Problem C Safety Path

Time limit: 1 second

Memory limit: 1024 megabytes

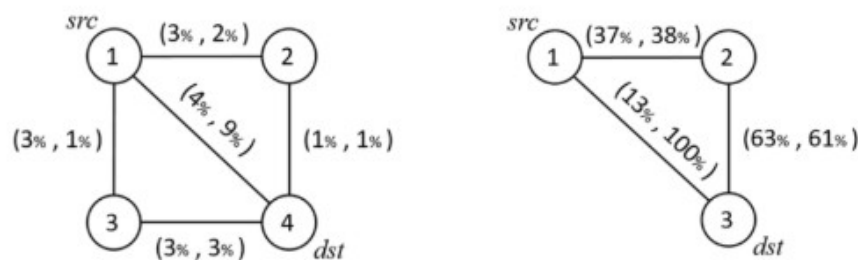
Problem Description

Goto Hitori is a freshman student in high school and she is an extreme introvert. She has troubles with most social interactions. When she was a junior high student, she always chose the path with the fewest persons, even though she made a detour.

However, on a fateful day, she coincidentally became the guitarist of the “Kessoku Band”, and she started to learn to interact with other persons. But, she still feels hard to interact with other persons. So, she will pass the crowd by consuming her “Stamina” and “Daily Social Energy”.

The top priority for Hitori is to select the route that consumes less stamina. If there exists more than one route consuming the same stamina, then Hitori will choose the one route with consuming the least social energy. Note that if Hitori runs out of stamina or social energy, she will pass out immediately.

The following figure is the information of the paths between buildings in the campus. The stamina and social energy will be labeled by (st, se) on each path, where st represents the stamina and se represents the social energy. For example, if Hitori chooses the path (v_1, v_4) with label $(4\%, 9\%)$. After passing the path, she will consume 4% stamina and 9% social energy.



In this case, the best route from building 1 to building 4 is $1 \rightarrow 2 \rightarrow 4$. Hitori will consume 4% stamina and 3% social Energy.

For another example, there does not exist any legal route from building 1 to building 3. The route $1 \rightarrow 3$ will deplete 100% social energy, and the route $1 \rightarrow 2 \rightarrow 3$ will deplete 100% stamina. Therefore, Hitori will pass out in this example.

Your task is to write a program to help Hitori to choose a safety route that she can go back home safely.

Input Format

Input consists of several test cases. The first line of each test case contains two positive integers, n , e , where $n(2 \leq n \leq 25)$ indicates the total number of street blocks, $e(1 \leq e \leq 300)$ indicates the

total number of paths between these n street blocks. The second line of each test case contains two positive integers, src, dst ($1 \leq src, dst \leq n$) indicates the source and the destination. Each of the next e lines contains four integers, x, y, st, se ($1 \leq x, y \leq n, 1 \leq st, se \leq 100$), representing that she could consume st percent of stamina and se percent of social energy from block x to block y . The input is terminated when both n and e are '0'.

Output Format

For each test case, output the remaining percentages of two energies in the best route. If Hitori cannot back to home safely, output "Passed out".

Technical Specification

- $2 \leq n \leq 25$
- $1 \leq e \leq 300$
- $1 \leq src, dst, x, y \leq n$
- $1 \leq st, se \leq 100$
- There is at most one edge between two street blocks.

Sample Input 1

```
4 5
1 4
1 2 3 2
1 3 3 1
2 4 1 1
1 4 4 9
3 4 3 3
3 3
1 3
1 2 37 38
2 3 63 61
1 3 13 100
0 0
```

Sample Output 1

```
96% 97%
Passed out
```